

System Logging Notes

I. System Logging Overview

What are System Logs?

- Services, kernel, daemons constantly generate data that gets saved as logs
- Provides human-readable journal of system events
- Logs are typically stored in `/var` directory (variable data location)

How Logging Works

- **syslog service** sends information to the system logger
- **syslogd** daemon (or **rsyslogd** on newer distributions) handles the process:
 - Waits for event messages
 - Filters messages based on importance
 - Routes messages to files, console, or discards them

Example Log Entry

```
Jan 27 07:41:32 icebox anacron[4650]: Job `cron.weekly' started
```

Format breakdown:

- `Jan 27 07:41:32` - Timestamp
 - `icebox` - Hostname
 - `anacron[4650]` - Service and process ID
 - `Job 'cron.weekly' started` - Event description
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2. syslog Configuration

Modern Implementation

- **rsyslog** is the advanced version of syslog
- Used by most current Linux distributions
- Configuration files located in `/etc/rsyslog.d/`

Configuration Example

From `/etc/rsyslog.d/50-default.conf`:

```
# First some standard log files. Log by facility.
auth,authpriv.*                /var/log/auth.log
*.*;auth,authpriv.none         -/var/log/syslog
#cron.*                        /var/log/cron.log
#daemon.*                      -/var/log/daemon.log
kern.*                          -/var/log/kern.log
#lpr.*                         -/var/log/lpr.log
mail.*                          /var/log/mail.log
#user.*                        -/var/log/user.log
```

Configuration Format

- **Left column:** Selector (what to log)
- **Right column:** Action (where to send logs)
- **Important:** Not all applications use rsyslog

Manual Logging

Test logging with the logger command:

```
logger -s Hello
```

This entry will appear in `/var/log/syslog`

3. General Log Files

Primary Log Files

`/var/log/messages`

- Contains all non-critical and non-debug messages
- Includes: bootup messages (dmesg), auth, cron, daemon logs
- **Use case:** General system activity overview

`/var/log/syslog`

- Logs everything except authentication messages
- **Use case:** Debugging system errors and troubleshooting

Usage Strategy

These two files provide comprehensive system monitoring and are usually sufficient for most troubleshooting scenarios.

4. Kernel Logging

`/var/log/dmesg`

- **Content:** Kernel ring buffer information from boot time
- **Includes:** Hardware drivers, kernel info, boot status
- **Behavior:** Gets reset on every boot
- **Use cases:** Hardware issues, boot problems
- **Alternative access:** Use `dmesg` command

`/var/log/kern.log`

- **Content:** Kernel information and events
 - **Includes:** dmesg output plus ongoing kernel events
 - **Use cases:** Comprehensive kernel troubleshooting
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5. Authentication Logging

`/var/log/auth.log`

- **Content:** System authorization logs
- **Includes:** User logins, authentication methods

Example Entry

```
Jan 31 10:37:50 icebox pkexec: pam_unix(polkit-1:session): session opened
for user root by (uid=1000)
```

Analysis:

- User with UID 1000 opened a root session using pkexec
 - Authentication method: pam_unix
 - Useful for security auditing and login troubleshooting
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6. Log Management with logrotate

The Problem

- Log files consume significant disk space
- Need to balance log retention with storage efficiency
- Want to prioritize recent logs over old ones

The Solution: logrotate

- **Purpose:** Automated log management utility
- **Execution:** Typically runs daily via cron
- **Configuration:** Files located in `/etc/logrotate.d/`

Features

- Specify number of logs to retain
- Compress old logs to save space
- Automatic rotation based on size/time
- Customizable retention policies

Alternative Tools

While other log rotation tools exist, logrotate is the most commonly used and widely supported solution.

Key Takeaways

1. **Log Locations:** Most important logs are in `/var/log/`
2. **Primary Files:** `/var/log/syslog` and `/var/log/messages` for general troubleshooting
3. **Specialized Logs:** `/var/log/auth.log` for security, `/var/log/dmesg` for hardware
4. **Management:** Use logrotate for automated log maintenance
5. **Testing:** Use `logger` command to test log functionality

6. Configuration: Check `/etc/rsyslog.d/` for logging rules